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Evaluating visual detectors without ground truth

Data lineage and human review in helipad detection in satellite imagery of São Paulo

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Contents

ABSTRACT 3

1 RESEARCH QUESTION 3

1.1 Context and problem 3

1.2 Central question and subquestions 5

2 STATE OF THE ART 6

3 RATIONALE 7

4 OBJECTIVES 7

5 HYPOTHESES 8

6 THEORETICAL FRAMEWORK 8

6.1 Evidence without ground truth 8

7 METHODOLOGY 9

8 TIMELINE 11

PRELIMINARY BIBLIOGRAPHY 12

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Program: Technologies of Intelligence and Digital Design (TIDD), PUC-SP.
Structure: follows the program's project outline (SANTAELLA, [n.d.]).
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Responsible-use notice. Automated visual detection based on imagery is subject to errors and is not equivalent to operational, legal, safety, or aeronautical validation. This research evaluates detectors, not sites: it does not produce, nor does it aim to produce, any validation of helipads for use, landing, delivery, or regulatory compliance.

Abstract

Object detectors trained on few examples are often evaluated on a small validation set annotated by the team itself; these metrics alone do not characterize behavior in a real, unannotated region. In the case studied, Helipad Detector, four experiments with mAP@50 ranging from 0.993 to 0.994 yielded detection rates of 9.6% to 21.0% on the same 7,767 region × tile pairs in the field. The dissertation asks how much of this difference is explained by data lineage, validation, and artifacts, and what the precision of these detections is according to sample-based human review. It proposes a reproducible protocol combining a lineage audit, a controlled comparison with spatial splitting, and precision estimation through human review, treated as an act of image interpretation rather than ground truth. A study of reviewers' trust remains an extension conditional on time and ethics approval.

Keywords: object detection; model evaluation; human visual reference; image interpretation; trust in automation; satellite imagery.

1. Research question

1.1 Context and problem

Helipad Detector is a team-developed project with a public repository and a record of each author's contributions (MINDFUL AI RESEARCH, 2026a). It uses YOLO networks trained for a single class ("helipad"), applied to satellite image patches (256 × 256-pixel tiles, zoom level 19), with field evaluation in ten regions of Greater São Paulo, a scope retained by this research.

The project originated as coursework for the fifth-semester Neural Networks course in the undergraduate Data Science and Artificial Intelligence program at PUC-SP, taught by Prof. Rooney Ribeiro Albuquerque Coelho. The applicant has participated since June 2026. Her contributions underpinning this research are the automation of tile collection and geocoding, execution of experiments exp2, exp3, and exp4, and field evaluation across ten regions, expanded from the region explored by Pedro Vyctor Almeida, author of the original annotated dataset and exp1. The project's other deliverables (interactive dashboard, presentation, and reports in Portuguese and English) are in the Helipad Detector repository and are outside this research's scope (MINDFUL AI RESEARCH, 2026b, 2026c, 2026d, 2026e).

Ground truth is independent confirmation, on site or through an official source, of what exists at a detected location. **Human visual reference** is the label people assign from the same images: an act of interpretation, subject to ambiguity, and not equivalent to ground truth. The project lacks the former at scale.

The YOLOv8n and YOLO11n experiments are virtually indistinguishable on validation (mAP@50 of 0.993–0.994; Table 1). On the same 7,767 region × tile pairs (5,540 distinct tiles) across ten regions, at a confidence threshold of 0.25, they yield detection rates of 9.64%–21.00% (Table 2 and Figure 1): the highest is more than twice the lowest.

Table 1. Validation-set performance (best epoch by mAP@50–95)

Exp.	Model (epochs)	Data (train / val.)	Best epoch	Prec.	Recall	mAP @50	mAP@50–95 best (last)
exp1	YOLOv8n (60)	Project-2 (178 / 34)†	54	1.000	0.963	0.994	0.881 (0.841)
exp2	YOLOv8n (100)	Project-2 (178 / 34)	81	0.982	0.971	0.993	0.888 (0.841)
exp3	YOLOv8n (100)	Project-1, augmented§ (356 / 34)	78	0.943	1.000	0.993	0.885 (0.841)
exp4	YOLO11n (100)	Project-2 (178 / 34)‡	54	0.970	0.948	0.993	0.877 (0.854)

Source: experiments' results.csv and args.yaml in the repository; seed 42 and 640-px images in all experiments.

†Attribution based on documentary evidence, not a count record: the preserved args.yaml and weights for exp1 (Pedro Vyctor Almeida's "exp1-2" run) point to dataset version 2 and are byte-for-byte identical to the applicant's copy; the training record saved in the notebook (116 training and 33 validation images) is compatible with the original exp1 run on version 1, of which only args.yaml remains. Strong evidence, to be confirmed in Stage 1. ‡Same data as exp2, but Ultralytics 8.4.163 (exp2: 8.4.98), which flagged three training label files mixing polygons and boxes as corrupt. §A fork of the dataset with augmentation (±15° rotation, brightness, and flips).

Thus, no comparison between the repository's existing experiments currently constitutes a controlled experiment; exp2 × exp4 comes closest, subject to the different Ultralytics version and the three flagged label files.

Table 2. Field detection (confidence threshold 0.25; 7,767 region × tile pairs, ten regions)

Experiment	Tiles with detections	Rate	Wilson 95% CI (naïve)
exp1	749	9.64%	9.01–10.32%
exp3	1,389	17.88%	17.05–18.75%
exp4	1,423	18.32%	17.48–19.20%
exp2 (rerun)	1,631	21.00%	20.11–21.92%

Source: repository detection_summary_by_region*.json reports. The production run of exp2 (7,943 tiles) detected 1,677 (21.1%). The naïve interval treats tiles as independent and therefore underestimates uncertainty, because neighboring tiles are spatially dependent.

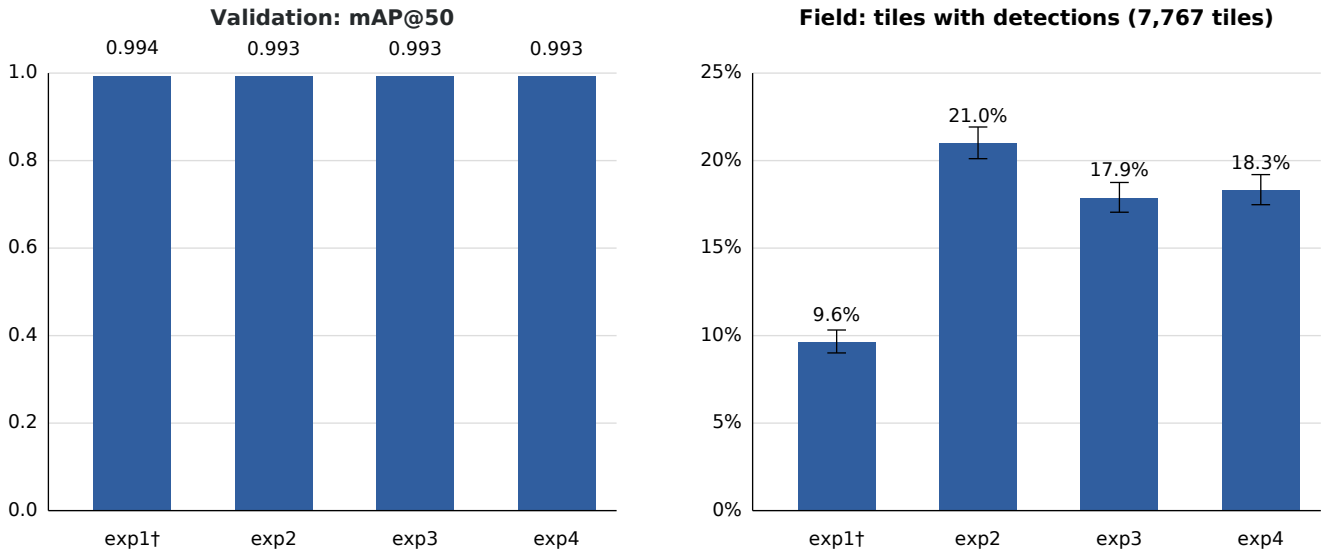


Figure 1. Same model family, nearly identical validation metrics, and very different field detection rates. Error bars: naïve Wilson 95% interval. †Dataset version for exp1 attributed through documentary evidence (Table 1). Source: author's elaboration using repository data.

The detection rate is the fraction of tiles with at least one detection; it is neither precision nor recall. Without ground truth, a higher rate cannot be interpreted as either more helipads found or more false positives. The project currently has only case-by-case reviews selected without random sampling (in Faria Lima, for example, three correct detections, two false positives, and an empty tile caused by a download failure and marked as a detection). Thus, field precision and the share of the rate caused by defective tiles are currently unknown and will be estimated in Stages 1 and 3. An mAP close to 1 does not reveal this.

There is also a leakage risk, already partly measured. Pixel-by-pixel comparison of the annotated images with field tiles reconstructed from the repository (5,540 distinct tiles in 7,767 region × tile pairs, because neighboring regions overlap) shows that 125 of the 225 geolocated images (87 of 177 training, 25 of 34 validation, and 13 of 14 test images) are copies of a field tile, corresponding to 2.3% of distinct tiles. The remaining 100 do not match any field tile and lie outside the bounding boxes of the ten regions. Training has therefore already seen 87 field tiles, and the detection rate on these tiles tends to be inflated; the overall effect cannot be quantified without per-tile detections. There are no direct copies between validation or test and training, but 15 of the 34 validation images and 3 of the 14 test images are within one tile (about 70 m) of a training image. This proximity will be treated as a risk, not confirmed leakage.

1.2 Central question and subquestions

Central question. How much of the difference in field detection between YOLO detectors trained on few examples is explained by data lineage, validation choices, and data artifacts, and what is the precision of these detections according to sample-based human review?

Subquestions.

• **RQ1 (lineage audit).** How much of the field differences stems from data lineage (version, split, overlap with field tiles, defective tiles), and how much from architecture?

• **RQ2 (controlled comparison and spatial splitting).** How much do best-epoch selection on the validation set itself and random splitting overestimate performance in new regions?

• **RQ3 (sample-based human review).** What is field precision, by confidence band and region, according to the human visual reference, and how reliable is that reference (agreement and proportion of "uncertain" labels)?

The influence of confidence presentation on reviewers' decisions (RQ4, exploratory) is an extension conditional on time and ethics approval (section 7).

2. State of the art

Detection in aerial imagery. Object detection in remote sensing is reviewed by Cheng and Han (2016) and supported by large datasets such as DOTA (XIA et al., 2018; DING et al., 2022). The YOLO family (REDMON et al., 2016; ULTRALYTICS, 2026), pretrained on COCO (LIN et al., 2014), underpins the models used, while Padilla et al. (2021) show that mAP depends on implementation conventions that must be stated. These works address large annotated datasets, not reliance on a detector trained on few examples in an unannotated region.

Helipad detection. Helipad detection in aerial and satellite imagery has already been studied. Bitoun and Winkler (2020) present HelipadCat, a dataset of around 6,000 images in 12 categories built from the FAA database, and a detection method using Mask R-CNN, area and shadow filters, and map-marker generation. Specht et al. (2021) train convolutional networks (ResNet101, Inception-V3, Xception, and EfficientNet-b0) to classify satellite images as helipad or non-helipad, using 9,324 images, and report around 95% validation accuracy with EfficientNet-b0; the authors also note inaccuracies in the FAA 5010 database. Suprapto et al. (2021) use Tiny YOLO for onboard-camera helipad detection for drone landing, a task distinct from this research. These works show that helipad detection itself is not the contribution here, and their results are not comparable to this research: classification accuracy and onboard-camera detection are not equivalent to mAP or detection precision in satellite imagery. This research focuses on evaluating the detector in an unannotated region, using spatial splitting and sample-based human review.

Evaluation validity. Kaufman et al. (2012) formalize data leakage. Roberts et al. (2017) and Ploton et al. (2020) show, for spatial data, that validation respecting spatial structure reveals worse performance than initially apparent; these findings come from ecology, and their transfer to satellite-image detectors is a hypothesis to test (H2). Dietterich (1998) addresses tests for model comparison, and Brown, Cai, and DasGupta (2001) address proportion intervals, the basis of the Wilson intervals in Table 2.

Calibration. Guo et al. (2017) show that modern networks tend to be poorly calibrated in image classification; calibration of detectors with limited data will be investigated through precision by confidence band (Stage 3), as confidence is the main signal reviewers receive.

Image interpretation and cognition. In image semiotics, seeing is not recording a fact but interpreting a sign, and interpretation depends on conventions, background knowledge, and context (SANTAELLA; NÖTH, 1998). A satellite image is a representation produced by a sensor at a specific date and resolution; using it to assert what exists on the ground requires inference. Distributed cognition (HUTCHINS, 1995) describes reasoning as a process extending across people, tools, and procedures. Both support treating detection review as a cognitive activity situated in a digital environment, not as neutral reading of an answer key.

Trust in automation and transparency. Parasuraman and Riley (1997) distinguish use, misuse, disuse, and abuse of automation; Lee and See (2004) address appropriate trust; Amershi et al. (2019) propose human–AI interaction guidelines. Model Cards (MITCHELL et al., 2019) and Datasheets for Datasets (GEBRU et al., 2021) propose reporting intended uses and limitations, a practice adopted by this project.

Gap. In the preliminary review, limited to sources already consulted, no work was identified that combines, for a rare urban object in satellite imagery, controlled comparisons, spatial validation, and precision estimation through sample-based human review treating the human reference as interpretation, with agreement and uncertainty reported. The exploratory web search used

combinations of “helipad detection”, “satellite imagery”, “aerial images”, “deep learning”, and “YOLO”, and was neither systematic nor exhaustive. The systematic review (Stage 1) may change this conclusion.

3. Rationale

Theoretical and practical. The works reviewed address metrics, evaluation validity, calibration, and trust in automation separately. This case brings them together in a concrete problem and shows a gap between validation metrics and field behavior that additional training alone does not resolve. The resulting protocol (lineage audit, spatial splitting, sample-based human review, and audit log) can be reapplied to other detectors and regions, turning an uninterpreted detection rate into estimates with explicit uncertainty.

Alignment with TIDD. The object of this research, within Cognitive Processes and Digital Environments, is not the detector itself but the production of evidence about it in a digital environment: people view satellite images through an interface, assign labels (helipad, non-helipad, uncertain), and use them to decide whether to trust the system. Two cognitive phenomena are involved. First, the human visual reference is an act of interpretation: it depends on reviewers’ background knowledge and image resolution and date, generating disagreement and hesitation that must be measured, not eliminated. Second, reviewers’ trust in the detector is a human–automation interaction phenomenon (PARASURAMAN; RILEY, 1997; LEE; SEE, 2004), influenced by what the interface displays (AMERSHI et al., 2019). Alignment is also evident in the program descriptions: the concentration area addresses “modeling cognitive performances: perception, attention, inference”, and line 4 describes the transition to “a world of probabilistic machines, implying distinct logics and risks”, with attention to AI’s ethical and social impacts. A detector whose confidence is interpreted by people is a concrete instance of this world. Thus, the research belongs to line 4, through evidence about a probabilistic machine and the limits of automation, and secondarily to line 3, through human interaction with AI systems and governance.

Social and ethical. A helipad detector may be misinterpreted as evidence that a site is safe, operational, or authorized. None of this follows. All outputs will be treated as analytical and non-authoritative, carrying the notice at the start of this document.

4. Objectives

General objective. To propose and test a reproducible protocol for evaluating visual helipad detectors in satellite imagery without ground truth at scale, treating human review as an interpretive reference, with uncertainty and agreement reported.

Specific objectives.

1. Reconstruct the data and configuration lineage of existing experiments, checking training–field overlap and defective tiles (RQ1).

2. Conduct controlled comparisons between YOLOv8n and YOLO11n and quantify the optimistic bias from best-epoch selection and random splitting relative to a spatial split by region (RQ1 and RQ2).

3. Estimate field precision by confidence band and region through blinded, sample-based human review by two reviewers (the applicant and Pedro Vyctor Almeida), reporting inter-reviewer agreement and the handling of “uncertain” (RQ3).

4. Document the system and data with a Model Card, Data Card, and auditable review protocol.

Expected outcomes. A lineage report; performance estimates with uncertainty intervals by split and region; a precision estimate through sample-based human review; a Model Card, Data Card, and review protocol; at least one article. No operational, legal, or aeronautical validation of any detected site is expected or will be claimed.

5. Hypotheses

Hypothesis	Test	Refutation
H1. With virtually equal validation mAP, field detection rate varies more across dataset versions and artifacts (such as empty tiles) than across architectures trained on identical data.	Stages 1 and 2	The difference between architectures, using identical data and multiple seeds, is greater than the difference between dataset versions.
H2. Best-epoch selection on the same validation set and random splitting overestimate performance; metrics decline measurably under spatial splitting.	Stage 2	The difference between random and spatial splits lies within the uncertainty interval (bootstrap).
H3. Field precision estimated through human review is lower than validation precision; variation between regions will only be described.	Stage 3	The 95% interval for field precision is compatible with validation precision. With about 30 items per region, regional differences will not be tested, only described.

Extension (H4, exploratory). Displaying calibrated confidence and examples of typical errors reduces reviewers' acceptance of false positives compared with displaying only the box and raw confidence. Its expected size will not support strong inference.

6. Theoretical framework

The research rests on four axes selected because they arise from the gap between validation metrics and field behavior. The first is **detection as probabilistic prediction**, summarized by precision, recall, and mAP (LIN et al., 2014; PADILLA et al., 2021). The second is **evaluation validity**: leakage (KAUFMAN et al., 2012), spatial dependence (ROBERTS et al., 2017; PLOTON et al., 2020), and model selection on the reporting set itself are sources of optimism. The third is **image interpretation and the notion of evidence** (6.1). The fourth is **trust, uncertainty, and responsible AI**: calibration (GUO et al., 2017), appropriate trust (PARASURAMAN; RILEY, 1997; LEE; SEE, 2004; AMERSHI et al., 2019), transparency (MITCHELL et al., 2019; GEBRU et al., 2021), and personal data protection (BRASIL, 2018), as rooftop imagery can enable inferences about people and buildings.

6.1 Evidence without ground truth

Interpreting an image is a situated act: the same patch may be read differently depending on the viewer's background knowledge and the image's resolution and date (SANTAELLA; NÖTH, 1998). The reviewer's judgment forms within a system that includes the interface, criteria guide, and model output (HUTCHINS, 1995). The human label is therefore not treated as an answer key. **Evidence** here means any claim about the detector's field performance accompanied by traceable provenance (lineage), a reproducible procedure, stated uncertainty, and an indication of its level (Table 3). A claim without a declared level is not evidence. Inter-reviewer disagreement and the "uncertain" category are study data, not noise to discard.

Table 3. Evidence levels adopted in this research

Level	Definition	What it supports	What it does not support
1. Validation metric	Precision, recall, and mAP on the team-annotated set.	Performance on the small validation set.	Performance in unannotated regions.
2. Field detection rate	Fraction of tiles with at least one detection.	Comparing detectors on the same tiles.	Precision, recall, or helipad count.
3. Sample-based human visual reference	Labels from two reviewers in a stratified sample, with agreement and “uncertain”.	Estimating precision by stratum, with explicit uncertainty.	Ground truth; large-scale recall.
4. Ground truth	Independent confirmation, on site or through an official source.	Outside the scope.	Will neither be produced nor claimed.

7. Methodology

Research type. Applied and empirical, predominantly quantitative (computational experiments and statistical estimation), with an interpretive component: human review as the production of a visual reference.

Starting material and authorization. The 3-project-ai-ml-yolo-helipad_detector repository, under Mindful-AI-Research, an organization created by the applicant, is public and contains experiments exp1–exp4, field reports, and documentation. Use of the repository and data for this research is authorized by Pedro Vyctor Almeida, coauthor of the original annotated dataset, in an authorization agreement signed by Pedro Vyctor Almeida and the applicant. The dataset is small (226 images in version 2: 178 training, 34 validation, and 14 test), with imagery from a single provider and date. The provider’s terms for derived datasets and redistribution will be checked in Stage 1 before any data publication. The project presentation summarizes the work and method.

Table 4. Dependencies and alternative plans

Dependency	Risk	Plan B
Dataset version for exp1	No count record corresponding to the preserved metrics; version 2 attributed through documentary evidence.	If Stage 1 cannot confirm it, exp1 is excluded from comparisons and retained only as a descriptive observation; RQ1 and RQ2 do not depend on it (Stage 2 retrains).
Flagged labels in exp4	Three training label files mix polygons and boxes; Ultralytics 8.4.163 flags them as corrupt.	Stage 1 standardizes labels; Stage 2 retrains both models on the same version using verified data; if infeasible, exp2 × exp4 is reported with a caveat.
Authorization to use material	Pedro Vyctor Almeida withdraws the signed authorization, or the third-party provider’s terms (Esri World Imagery) restrict use or redistribution of images and the derived dataset.	Restrict use to material authored by the applicant (collection, geocoding, exp2–exp4, field reports), reannotating as necessary.
Second reviewer	Pedro Vyctor Almeida has agreed (signed agreement); the risk is unavailability during the project.	Invite a substitute with a similar profile; without one, report results from a single reviewer and state this limitation.

Stage 1. Lineage and reproducibility audit. Reconstruct each experiment’s dataset version, split, configuration, and seed; confirm exp1’s dataset version; standardize labels (most files contain polygons converted into boxes by detection training; three mix polygons and boxes); and check the effect of the Ultralytics version. Extend the existing training × field cross-check (tile copies and proximity) to missing Faria Lima tiles and each experiment’s per-tile detections to measure overlap, measure the fraction of empty or corrupt tiles sent to inference, and pin data, code, and dependency versions. In parallel, conduct a systematic literature review including image semiotics and cognition.

Stage 2. Controlled comparisons and spatial splitting. Train YOLOv8n and YOLO11n using the same data, split, and epochs, with at least three seeds, and compare using paired-model tests such as McNemar and 5×2 cross-validation (DIETTERICH, 1998), plus Wilson intervals (BROWN; CAI; DASGUPTA, 2001). Then train and evaluate with region-based splitting (leaving one region out each round, where feasible), compare against random splitting, and quantify the difference. Because neighboring regions share tiles (5,540 distinct tiles in 7,767 pairs), splitting will use distinct tiles, removing from training any tiles shared with the held-out region. Baixada Santista is a candidate reserve region, provided its absence from training is confirmed through coordinates; Faria Lima is suitable only if the two training images already matching its tiles are excluded. The epoch will be selected on a set separate from the reporting set.

Stage 3. Sample-based human review in the field. Randomly sample, stratified by confidence band and region, detections from the Stage 2 model(s) on the 7,767 common region × tile pairs, plus tiles without detections to estimate omissions, and produce the human visual reference following Table 5, with two reviewers: the applicant, Fabiana Rotella Campaner, and Pedro Vyctor Almeida. Precision by stratum is estimated using Wilson intervals and spatial block bootstrap; the confidence threshold is varied to show changes in precision and coverage, and precision by band feeds the reliability diagram (GUO et al., 2017). The result is a human visual reference, not ground truth.

Table 5. Human review protocol (Stage 3)

Item	Definition
Reviewers	Two: the applicant and Pedro Vyctor Almeida, who has agreed to participate. He annotated the original dataset and trained exp1, so he is not independent of the project; this limitation is declared, and an external reviewer will assess a subsample if available.
Independence and blinding	Each reviewer works alone. A script generates the queue with a fixed seed in random order, mixing detections and tiles without detections; reviewers do not see confidence, source experiment, or stratum. Both know the detector, and the second reviewer annotated the training data (his criteria may match the training-label criteria), so the main analysis is repeated separately for each reviewer (sensitivity analysis).
Categories and pilot	Helipad, non-helipad, and uncertain, with visual criteria in a written guide. A pilot of around 50 items adjusts the guide and sample and calculates agreement (Cohen’s kappa).
Sample (order of magnitude)	Three confidence bands (0.25–0.50; 0.50–0.75; ≥0.75), with around 100 detections per band randomly sampled with a minimum quota per region; strata with fewer than around 100 are fully reviewed. Plus around 100 tiles without detections. Total: approximately 400 items per reviewer, to be adjusted after the pilot.
Power for rare strata	With n ≈ 100 per band, the Wilson interval half-width is about ±10 percentage points (p = 0.5), and comparing two bands has around 80% power for a 20-point difference (50% versus 70%). Per region (n ≈ 30), the half-width reaches ±17 points, and results are descriptive. With few expected positives, only the upper bound is reported (rule of three: zero in 100 implies about 3%). Spatial dependence widens intervals, so block bootstrap is used.
“Uncertain” and disagreements	“Uncertain” is a separate category, never silently merged: precision is reported treating “uncertain” as non-helipad, as helipad, and as excluded, alongside its proportion by stratum. Kappa is calculated before resolution; disagreements are discussed in a recorded session, and unresolved items remain “uncertain”.

Stage 4. Documentation and audit. Produce a Model Card, Data Card, review protocol, and audit-log schema, with versioned artifacts.

Conditional extension (H4, RQ4). Compare two interfaces (box and raw confidence versus calibrated confidence with examples of typical errors), measuring false-positive acceptance and decision time. Ethics approval is required: the project will be submitted in semester 2, and the extension will run only if approval is received before semester 4 and time permits; otherwise it remains future

work, without affecting the core scope. Study design and analysis methods will be explored further in a human–computer interaction course (section 8) and in the works listed for future consultation.

Ethics, privacy, and limitations. No publication will identify specific people or buildings, and results will be aggregated by region; any need to process personal data will be assessed under the LGPD (BRASIL, 2018). Expected limitations: a small, geographically heterogeneous dataset; a single provider (Esri World Imagery) and image date; lack of ground truth at scale; human review costs; visual interpretation ambiguity; and limited power in rare strata.

8. Timeline (24 months, within the program's 18–30 months)

Semester	Activities	Deliverables
1 (months 1–6)	Courses: Epistemology and Research Methodology (EMP, required) and Foundations of Artificial Intelligence (FIA, line 4, module 1); foreign-language proficiency within 6 months of admission (admissions notice, item 6.4); systematic review; Stage 1; review protocol design.	Lineage report; review protocol (v1)
2 (months 7–12)	Courses: two electives, with Human–Computer Interaction and User Experience (line 3) and Theory of Signs, Symbols, and Codes (line 4) as candidates; Stage 2; review pilot; submission to the ethics committee (extension).	RQ1 and RQ2 results; pilot report
3 (months 13–18)	Qualification examination (semester 3); Stage 3; decision on the extension, with ethics approval in hand.	Qualification document; RQ3 results
4 (months 19–24)	Stage 4; reviewer extension if approved and time permits; writing; article; defense.	Dissertation; at least one article submitted

The TIDD Master's program lasts 18–30 months (program webpage); 24 months are proposed. Candidate courses depend on availability, as the program states that each course syllabus receives a specific content description each semester; qualification follows the program regulations.

Preliminary bibliography

Preliminary list, following ABNT standards, of works consulted in preparing this proposal, followed by works to consult. It will be expanded during the systematic literature review.

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